

ARMY GENERAL



PHOBETOR



English manual containing the rules for version 1.00

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1. Introduction

Army General is a turn based strategy game that deals with campaigns that took place in northern Africa during the second world war.

2. User interface

2.1. Symbols

Many menus contain symbols to deliver information. They are used as they use up fewer space compared to texts. If you hover over the symbols with the mouse cursor a tooltip will show up. Within it the meaning of the symbol will be explained.

2.2. Buttons

Buttons contain symbols or text in them that describe their function. Additional tooltips will provide more detail or subsequent information about possible actions.

2.3. Menus

2.3.1. Main menu

The main menu appears automatically after loading the game and offers the following options:

Campaign

The player can choose one of the two campaigns in the next menu (See 18. Campaigns)

Scenario

Opens the scenario selection. The player can choose which side he wishes to fight for and select rules and difficulty settings.

Last

Automatically loads the last savegame.

Load

Savegames of campaigns or scenarios can be loaded in the follow-up menu

Run Editor

Runs the editor of Army General. See also chapter 19.

Options

General settings of the game can be adjusted here. See chapter 4 for detailed information.

Credits

Shows who worked hard to complete Army General.

Quit game

Return to the desktop.

2.3.2. Start a campaign or a sceanrio

The player can adjust the rules and the difficulty for most scenarios or campaigns to his liking in the selection screens for campaigns or scenarios. The following options are available: :

- **Weather:** Influences combat and movement if activated; generally speaking bad weather is reducing combat effectiveness and range of units. The exact influences can be found in chapter 15.
- **Fog of war:** If this option is active each player can only see those units of the enemy that are within the view of his own troops. If it is deactivated both sides can see all units all the time.
- **Supply:** With this rule activated units have to be supplied with ammunition and fuel from time to time. This is using up 2 action points of a unit. For most units this means they cannot perform any other action during that turn. Without this rule, units have unlimited fuel and ammunition.
- **Boni:** Each side can get up to 1500 additional resources per turn and up to 1500 experience points that are giving each unit once.

3. The Panel

After a scenario has begun there is the panel on the left side. Important information on the landscape and the units are displayed here. Buttons for certain actions and others that lead to sub-menus are also placed there.

3.1. General information and end of turn



Right at the top there is a display of the weather conditions - on the left the current weather on the right the forecast for the next turn. In between those displays there is the flag of the nation of the current player.

Below it there are the players resources shown on the left. Next to it is the button that ends the turn (the consequences can be studied in chapter 17. End of turn). Right of it the current turn number and possibly a time restriction is displayed.

3.2. Buttons



The next area contains eight buttons that partly lead to sub-menus and partly influence the game directly. This list is explaining the function of the buttons leading from left to right and top to bottom:

- **Undo move:** Clicking on it will undo the last movement of the player. It will not work if the player has spotted new enemy units.
- **Next unit:** The game jumps to the next unit with action points.
- **Air/ground display:** Toggles between air and ground view.
- **HQ:** Lead the player to the headquarters where you can recruit new units and rearrange the existing ones. See also chapter 14.
- **Quick save:** Saves the game using an automatically generated name instantly.
- **Load/Save/Restart/Quit game:** In this menu one can load or save a game, restart the current mission or quit the mission. This can be done by returning to the main menu or by quitting the game completely.

- **Options:** All general settings for the game are adjusted in this screen. For more detail see chapter 4.
- **Objectives and statistics:** Statistics, combat protocols, objectives and general hints for the game can be viewed by the player here.

3.3. Display of the attributes of the units



After a unit was selected its attributes will be shown in the middle of the panel. If the selected unit is owned by the player, there will be additional buttons to refit the unit. From left to right and from the top to the bottom the following information is available:

- **Officers picture:** Graphical depiction of the officer.
 - **Unit name:** Shows the current name of the unit, but as it is also an input field it can be changed by the player any time by clicking into the area.
 - **Core unit:** If the unit is a core unit during a campaign, a golden badge is displayed.
 - **Unit type:** Stands for the combat type that is used to determine which values are used by the enemy against it.
 - **Rank and name:** Each officer has a personal mark. Ranks may change in the course of a game.
 - **Leadership:** This value is responsible for the amount of regeneration of the combat readiness of a unit caused by the officer. See 13.1.
 - **Combat readiness:** This value shows the percentage of the unit's effectiveness. A value of 100% equals the full power of e.g. an attack value. Movement, battles and being bombed by the enemy reduces this value, but it never drops below 15%. The combat readiness is restored by resting and by its officers leadership.
- Example: A unit with an attack value of 10 has a combat readiness of 50%. In battle this results in an attack value of 5.
- **Experience:** The experience points the unit has gathered so far..

- **Unit modifiers:** Units have advantages and disadvantages which depend on the circumstances e.g. tanks are handicapped in swamps. The symbols for these modifiers are displayed here. Use tooltips to get more details and see chapter 10.2.
- **Unit image:** Depiction of the main equipment of the unit as seen on the map.
- **Combat values:** In the upper row are the attack values, in the middle are the respected symbols and in the lower row are the defence values for each combat type. Those are from left to right: Infantry, tank, cannon, aircraft and ships. Those are basic combat values and they are altered by various factors. See chapter 10. Battles.
- **Designation:** Exact type name of the units' main equipment
- **Support units:** Up to two sub-units are displayed here. Tooltips provide detailed information on the impact of those units. See also chapter 12.
- **Refit area:** Buttons for "Blitz supply" (refill fuel and ammunition for resources but without spending the units action points), "supply only" (no strength points will be added), "Reset", "set all to maximum" and "Apply". Below these buttons the current amount of ammunition and fuel as well as the costs of the current selection are shown.
- **View:** How many hexes can be seen by the unit. This is influenced by objects.
- **Weapons range:** This is the maximum distance the unit can attack an enemy.
- **Entrenchment:** Shows the maximum level of entrenchment that is possible for the unit.
- **Action point:** Current and maximum amount of action points
- **Movement points:** Current and maximum amount of movement points. Can be influenced by transports.

In general all values and icons have more information revealed if you hover over them with the mouse and wait for the tooltip to appear.

3.4. The mini map



In the bottom of the panel there is the mini map. It shows the whole map of the scenario miniaturized for a better overview. If you click in the mini map the display of the big map will move to that location.

Right of the map there is a column of buttons that control the symbols that are shown on the mini map:

- Hide/Show ground units
- Hide/Show air units
- Hide/Show mission objectives
- Hide/Show view areas of units
- Hide/Show deploy hexes
- Hide/Show camera view

4. Options



During a scenario you can get to this screen by clicking on the button with the crossed hammer and wrench. In the main menu there is a button titled "Options" which leads to the exact same screen, content and options are identical.

4.1.1. Screen Resolution

With this drop down menu you can choose between several resolution in which the game should run. The available choices depend on graphic card and monitor.

4.1.2. Full screen

Army General can be played in fullscreen or windowed mode. Besides toggling it on this occasion one can also press F-11.

4.1.3. Combat info

After a battle a combat report can be shown. If and how detailed it is displayed is determined by the selections "off", "reduced", "normal" and "detailed". All such reports are stored in the combat protocol of a map regardless of choice made here. (See 9.6. Combat results)

4.1.4. Display unit movement

If switched on, the movement of the units during the players turn is shown. The speed of this animation can be adjusted with the sliders discussed in 4.1.15. If this option is deactivated, the units will simply appear on the destination hex.

4.1.5. Display unit movement of AI

If the AI makes its turn, you can watch it move and attack on the map. As this is prolonging the time of its move, you can switch the animation of that part on and off.

4.1.6. Show combat results of the AIs' attacks

If this is activated the combat results of the AIs' attacks will be shown during the turn of the AI.

4.1.7. Center on selected unit

This causes the camera to center on the selected unit. If you select a unit that is originally on the border of the screen, the camera will rush to the unit and it will be seen in the middle of the screen.

4.1.8. Combat effects

If units engage in combat there are battle animations to depict the fight. With this option you can turn them on and off.

4.1.9. Weather

Weather will not only be simulated with values, but also displayed on the map if this option is active.

4.1.10. NATO symbols

You can choose if you want to play with NATO symbols or figures to represent the units on the map. If you choose this option you will always see the NATO symbols in both zoom levels. If you deactivate it, you will see figures in the near view and NATO symbols if you zoom out.

4.1.11. Create a savegame at each turn

If this is selected the game will create a savegame at the beginning of each new turn. Beware that this might create many savegames over time and may cause a delay if they all need to be listed while loading a savegame.

4.1.12. Language

Army General is available in German and English. To switch to the other language you have to choose it in the drop down menu. If the new language is now seen in the display you have to quit the game and restart it to finish the change.

4.1.13. Volume

With the six sliders bearing the term "volume" at the top and you can mix the sound of the game to your liking. The buttons on the left turn each sound channel on and off.

4.1.14. Zoom speed

You can choose two zoom levels of the map with the mousewheel or with the page up & page down keys. This slider adjusts how fast it happens.

4.1.15. *Speed of movement*

This slider controls the speed with which units move from their starting hex to their destination. To see any result, the animations have to be switched on, see 4.1.4.

4.1.16. *Hex field grid*

To show the structure of the hexes in the landscape the game creates a border for each hex. The slider controls the density of that line.

5. The main map

The main map which is the scene for the missions is a 2d map with two zoom levels which can be toggle with keys or the mouse wheel. Besides the units there is terrain and objects on the map.

5.1. Terrain

The terrain influences the view of units and consumes different amount of movement points depending on the nature of it. Also there is a maximum level of entrenchment that varies from terrain to terrain. Units cannot have a higher level of entrenchment than the terrain allows them to. The details are listed here:

Terrain	Movement	Sicht	Verschanzung	Besonderes
River	4	1	0	Impassable for ground troops except for infantry, units with transports need 2 additional movement points.
Mountain	-	4	4	Impassable for all ground units except for mountain troops.
Lake	-	1	0	Impassable for ground units
Forest	3	2	3	
Plain	2	1	2	
Swamp	3	1	1	
Hill	3	2	3	
Street	1	1	2	
Railroad	-	1	2	Allows operational movements in between linked cities
Desert	2	1	2	
Sea	1	1	0	

5.2. Objects

On the map there are objects which have an owner. But the owner can change if an object has been occupied by the opponent. The only exeption are minefields. They cannot be conquered, only be removed. Objects can influence movement and view of units. An object also scouts the adjacent hexes for its owner. A flag on the hex of an object shows the current owner of it.

5.2.1. Boni

If an object is conquered the first time during a mission, it might happen that the conquerer receives a bonus once. The distribution of such a bonus depends on the scenario design.

Possible are those boni:

- **Units:** The player receives core or auxiliary units that either appear on the map or are added to the reserve.
- **Ammunition:** The conquering unit gets its ammunition restocked. der Einheit, die das Objekt erobert hat, wird der Munitionsvorrat aufgefüllt
- **Fuel:** The conquering unit gets its fuel restocked.

- **Resources:** The player receives the noted amount of resources added to his resource pool.
- **Officers:** The player receives one or more officers added to the officers corps. They can then be assigned to units.

5.2.2. Visibility

Each object can be defined as invisible for one side. This means the object has to be discovered on the map by moving a unit onto or next to it. This makes e.g. sense if there are hidden minefields or depots that shall surprise the player. Some objects can only be discovered by ground units.

5.2.3. Overview:

Object	Movement	View	Entrenchment	Notable
dense urban area	2	2	4	Can be marked as supply center for ground units, can be part of a rail network
light urban area	2	2	3	Can be marked as supply center for ground units, can be part of a rail network
Minefield	2	1	2	Can have strength up to 10 points, damages enemy units that move on that hex. The damage depends on the strength of the field. It cannot be conquered.
Oilfield	2	1	2	
Bunker	2	1	5	Can be build by engineers, can be destroyed by bombers and engineers
Fort	2	1	8	
Bridge	1	1	0	Can be damaged by bombers, partisans and engineers. Engineers can repair a damaged bridge.
prison camp	2	1	3	
Airfield	2	1	2	Can be marked as supply field for aircraft. Can be used to embarked units into air transports. Aircraft can only be reinforced here.
Depot	2	1	2	
Port	2	1	2	

5.3. Displays

5.3.1. Hexes

Many options are marked by colouring hexes or the border of a hex.



Unit can disembark from a sea transport



Deployment of air units possible



Deployment of ground units possible



This unit can be attacked



Path of the units movement

5.3.2. Symbols



Artillery unit is marked after support fire



Sonar contact in this hex



A sea unit is located beneath the air unit



Railroad is blocked

5.3.3. Multiple choices

There are situations in which two options for a mouse click are available. If this is the case the cursor will change if moved between the upper and the lower part of the hex and each form represents a different action. For example if there is a ground and an air unit on the same hex and an neighbouring unit like a ship can attack either of them Another would be an air unit that can be attacked on a hex with no ground unit. Then the player could attack or move on the hex.

5.4. Building, damaging and repairing infrastructure

In the game there are infrastructures with which the player can interact: Bridges, bunkers and railroads. To damage infrastructure a unit has to be on the same hex, needs the ability to perform the action, have ammunition and needs all its action points. The same is true about repairing infrastructure except you need resources instead of ammunition. Which unit can be used for each action can be studied in 6.4. Special units. Damaged infrastructure is marked on the map.

Besides the action points, the following costs are incurred:

Action	Costs
Damage	1 ammunition
Repair	20 resourcen
Build	50 resourcen

6. Units

6.1. Unit types

Basically there are three kinds of units that can each have different sub types: land, water and air units. These are the sub types:

Land	Sea	Air
Infantry, tank, cannon	Ship	Fighter, Bomber

6.2. Core units

Units that can be transferred by the player from scenario to scenario during a campaign are called core units and are marked with a golden badge in the unit information. In the headquarter you can apply a filter to only see your core units. All other units are auxiliary troops and disappear after a scenario. On single maps this distinction is of no consequence as none of the units are used afterwards.

6.3. Auxiliary troops

Auxiliary troops are units that are given the player to solve a scenario. They might be on the map from the very start, appear as reinforcements in a certain turn or be given as bonus to reward an achievement. Auxiliary troops can not be transferred to the next scenario. Therefore it is advisable to reinforce them only if absolutely necessary as all invested resources are lost at the end of the mission while they are saved in core units.

6.4. Special units

Some units have special abilities:

Unit	Abilities
Engineers	Damage/repair railroads and bridges, build and destroy bunkers
Assault engineers	Lay or clear minefields
Bomber	Damage railroads and bridges, destroy bunkers
Paratroops	air drop on hexes except on lake, sea or mountain hexes
Partisans	Damage railroads and bridges
Mountain troops	Enter mountain hexes
Carrier	Can supply aircraft with ammunition
Destroyer	"Sonar" detects submarines
Submarines	Invisible due to silent running, can be detected by sonar

6.5. Selecting units

A unit can be selected on the map by a left or right click of the mouse in the hex the unit is located. The unit is then marked with four small arrows that point at the unit. Can the unit move or attack an enemy the hexes offering the actions are coloured or marked as shown in 5.3 Displays.

6.6. Refitting units

6.6.1. Ground units

Supplying ground troops is hindered by adjacent enemy. An increasing number of neighbouring enemies is limiting the kind of supply that can reach the concerned unit:

Number of enemies	Possible supply
0-2	All
3	ammunition, fuel
4	ammunition
6	Supply not possible

6.6.2. Air units

Air units can only be supplied by a friendly airfield or an aircraft carrier. For this it has to be on the same or an adjacent hex.

6.6.3. Ships

Ships can only be refitted in ports. The ship has to be directly on the hex of the port and needs all action points.

7. Action points

To execute actions, a unit needs action points. The amount of action points depends on the mission it is for:

Action	Costs
Refill fuel/ammunition/Strength	2 points
Lay mines (4 Strength points/turn)	2 points
Clear mines (5 Strength points/turn)	all points
Movement	1 point
Build bunker	all points
Repair	all points
Attack	1 point

Attacks can only be execute once in a turn, even if there would be enough action points for another one. Movement can be executed as long as the unit has enough movement points to do so.

Clearing mines is a passive action. The unit with the ability simply needs to stay on the same hex and do nothing.

8. Moving units

After a unit was selected on the map the possible destination hexes are coloured around the unit. The terrain has a huge impact on the range. If fog of war is activated the hexes that are not scouted yet are darker. If you move onto such a hex there is the danger of an ambush. (see 10.3.)

Units with transports (infantry and cannons) have the additional colour cyan to mark the area in which the unit ends its movement embarked in the vehicles. The mouse cursor is also different and shows a spinning wheel. Being embarked during the turn of the enemy means that the first attack on such a unit will lead to a lowered defence value and no damage for the enemy.

Are all movement points used up then the unit remains on the hex until the next turn of the player. Only a retreat during attacks of the enemy might change that. (see 10.4. Combat results)

Only infantry units can enter river hexes without an intact bridge. If the infantry unit has transports it will need additional two action points to enter a river.

Mountain hexes can only be entered by mountain units.

8.1. Zones of control

A unit always has a zone of control (ZOC) that is reaching into all adjacent hexes. Entering a zone controlled by an enemy unit increases the costs of movement. Zones of control can overlap and thus increase the cost for entering the hex. E.g. if two enemy units border a hex, the costs caused by the ZOC double if the player wants to move on that hex. If you move into areas that are not yet scouted then ZOCs in the invisible areas can stop the unit earlier than expected by the original display.

8.2. Transports

Infantry, artillery, antitank and antiair units can be embarked into transports to achieve a higher range during movement. Selfpropelled weapon systems cannot use transports. Transports can be added in the headquarter before a unit is deployed.

There are three kinds of land transport:

Name	Costs	Combat modifier	Range	Fuel	Assignable to
Wagon	250	40	4	30	Artillery, AT- & AA gun
Truck	500	50	8	50	Infantry, Artillery, AT- & AA gun
Halftrack	700	65	8	45	Infantry, Artillery, AT- & AA gun

8.3. Air shipment

If infantry is placed on an airfield it can be embarked on an air transport. In order to do this the airfield has to be set by the designer to allow embarkment. The unit needs to have full actions points to embark and the player needs 5 resources to enable the process. If there is another airfield owned by the player within the range of the air transport, the unit can be disembarked in the same turn if there is not other unit on the destination. After the disembarkment the unit cannot execute any other actions. In case there is no other airfield in range, the unit will stay embarked

and can be attacked in this state during the enemies turn. In such a combat the values of the aircraft will be used for the combat results.

8.4. Sea transport

Ground units placed on a port can be embarked on ships. The unit will then turn into a transport ship and can be moved on sea hexes. To disembark the unit, it can either move onto a port or next to an empty beach. Embarkment or disembarkment cost both all action points of a unit. In addition to that there can be costs in form of resources depending on the scenario setting. The amount of resources are determined by the formation and the number of support units.

8.5. Rail roads and operational movement

Two friendly cities that are linked via undamaged railroads can be used to transfer troops between them. The destination city must not contain a unit and the unit to be send must be in the other city. After the transfer the unit can not execute any other action. This operational movement costs 10 resources.

Possible connections are marked by arrows.

8.5.1. Damaged rails

Rails can be damaged by partisans, engineers or bombers. This can cause an interruption of a line and thus prohibit the transportation of a unit. Only engineers can repair damaged rails. To do this, the engineer unit has to move onto the hex and needs full action points for the mission. No resources are needed.

Enemy units that are located on a railroad will also block the line. Any disturbance of a line will be marked by a red border and crossed out hex.

8.6. Parachute units

Paratroopers that are airborne can perform an air drop on any hex except for lake, sea and mountain hexes.

9. View

Units have a certain range of view that is influenced by the environment. Usually each hex costs one view point. If all points are used up, the following hexes cannot be scouted. These hexes are darkened. Each unit and object can scout at least with the range of 1. The exact costs and influences can be studied in 5.1. and 5.2.

10. Battles

To execute an attack, the attacker has to be able to attack his enemy (e.g. tanks can't attack air craft, but it works the other way around), the enemy has to be within weapons range and the attacker needs to have ammunition.

The hex of the defender is the reference point for distances and the terrain. This is true for unit modifiers (see 10.2.) and support fire by third parties (see 10.5).

A battle can lead to the loss of strength points and depending on the outcome to additional consequences (see 10.4.). In any case both participants lose one ammunition point. All losses are subtracted from the unit values after the battle.

To start an attack the player has to select one of his own units. If enemies can be attacked their hexes are marked by a red line around the border. If he now moves the mouse cursor over such an enemy it will change its shape to a combat preview (see image 10) that shows the officers of both units and a rough estimation of the losses if the battle is executed. On the left there are the predicted losses of the attacker, on the right the ones of the defender. Depending on the settings, there will be a preview of the combat report display on the screen, too. In this report the factors leading up to this result are listed. A click with the left mouse button will execute the combat. The final combat result may vary from the preview.



Abbildung 1: Combat preview on the map

10.1. Entrenchment

A unit that was moved or used for an attack is automatically digging itself in. By this it is increasing its current entrenchment level by one step each turn it is inactive up to the maximum possible value for this specific unit. The maximum value is also determined by the amount the terrain allows or by objects such as bunkers or dense urban areas.

Any attack on an entrench unit lowers the entrenchment level at least by one. Entrenchment levels increase the defense values of the entrenched unit. Exceptions can be found in 10.2.

Infantry moving onto a hex with a bunker immediately gets an entrenchment level of 2. If the unit stays on the hex, the value will increase each turn by one level up to level 5.

As soon as a unit is moved, it will lose all entrenchment levels.

10.2. Unit modifiers

Units have specific modifiers that have different influences. Most of them change combat values if a unit fights on a certain terrain or if special cases occur during combat. Modifiers are shown in the panel if a unit was selected among its other attributes below the officer (see 3.3.).

There are the following modifiers in the game:

City penalty for armoured vehicles

In dense urban areas attack/defense -7

In light urban areas attack/defense -3

City penalty for selfpropelled artillery

Light urban areas attack -1/defense -3

Dense urban areas attack -2/defense -5

Forest penalty for armoured vehicles

Attack onto forest hexes -7

Defense of a forest hex -3

Forest penalty for fighter

Attack on hills -4

Hill penalty for tanks

Attack on hills -8

Defense on a hill -4

Hill bonus for mountain troops

Attack on hills +1

Defense of a hill against ground troops +1

Swamp penalty for armoured vehicles

Attack on a swamp hex -9

Defense of a swamp hex -9

Hill bonus for artillery

Chance of hitting a target increases for artillery units due to the elevated position.

Attack from hills +2

Defense of a hill +3

Radio doctrine 1941-43

German tanks are equipped with radio and thus can use it for better combat coordination.

Attack/defense against tank units +3

Disruption

Decreases the combat readiness of an enemy unit after an attack, the amount varies between 5% and 15%.

Bunker bonus

Damage that a unit suffers decreases by 1 in bunkers or forts.

Bunker killer

Ignores up to two levels of entrenchment as well as the modifier "Bunker bonus".

Air shelter:

Infantry units profit from the various options to take cover in cities.

Defense if under air attack:

Light urban areas +3

Dense urban areas +5

Ski-Guerrilla

Units with a ski battalion during snow:

View +1

Snow fighter

Units with special equipment and training are adapting perfectly to the conditions in snow and ice.

Ignores effects caused by weather condition "snow"

Recon doctrine

Evades damages up to 2 strength points

Evasive manoeuvre

Evades damages up to 2 strength points

Sparse enemies

Verringerter Schaden wenn der Gegner weniger als 50% seiner maximalen Stärkepunkte hat.

German artillery doctrine

Support fire for friendly units under attack - effectiveness 50%.

British artillery doctrine

Support fire for friendly units under attack - effectiveness 40%.

Close air support

If at least one friendly unit is in a neighbouring hex coordinating an air strike on an enemy ground unit:

50% chance of an increased attack value of up to +7

Battlefield air interdiction

If at least one friendly unit is in a neighbouring hex coordinating an air strike on an enemy ground unit:

50% chance of reducing the enemies combat readiness by 5 - 20%

10.3. Combat modifier

The following factors influence the combat results:

- Attack value of the attacker against the enemies combat type
- Defense value of the defender against the combat type of the attacker
- Officers and their advantages
- Current strength of each unit
- Ammunition: Defenders without ammunition suffer additional losses
- Entrenchment level of the defender: Each level is multiplied by three and the result is added to the defense value. Example: a defender that is entrenched to level 2 receives additional 6 points to his defense value.
- Rivers: an attack from a river hex is multiplied by 0.75; the defense of a river hex by 0.8.
- Ambush: Is a unit moved onto a hex with a so far undiscovered enemy an ambush takes place that immediately results in a battle. In this case the moving unit is the defender and the defense value is multiplied by 0.7; the attack value of the ambushing unit is multiplied by 2.5.
- Readiness: Attack and defense values are multiplied with the respective readiness of each unit.
- Level of the unit: Each unit can reach level by gaining experience. The value will be multiplied with the attack or the defense value of each unit. The exact values for each level can be seen in chapter 11. Experience.
- Unit modifiers: Depending on the unit type and equipment several modifiers may occur. See also 10.2.
- Support units (SUs): These sub-units can influence combat in many ways. Most times their values are added to the combat value of the main unit. For details and special cases see 12. Support units.
- Counter fire: If an artillery unit is attacked, then it returns fire if it has ammunition and the enemy is in its weapon range. Counter fire is submitted to a penalty and will only have 30% of the usual combat value.
- Weapons range: Firepower of artillery or anti-air is decreasing with the distance to the target. Each hex beyond the adjacent one costs 10% of the combat value.
- Flanking: The defense value of the enemy can be reduced by placing more than one unit on neighbouring hexes. Now the defender has to secure more directions and thus weakens each direction accordingly. The value changes this way:
 - One unit: 100% (normal combat value)
 - Two units: 85%

- Three units: 75%
- Four units: 65%
- Five units: 55%
- Six units: (surrounded) 50%

10.4. Combat results

After a battle one of the five following events takes place:

- **Holds position:** The defender blocks the attack and remains on his position on the hex.
- **Retreat:** The defender tries to retreat onto an unoccupied hex with no adjacent enemy unit thus avoiding enemy zones of control. There is a 25% chance that the unit loses its transport, 1-3 ammunition or 5-10 fuel points. Are there only hexes left that are in enemy ZOCs and the unit has a readiness of 50% or more, it can retreat on or through such hexes but will suffer 10% readiness per enemy ZOC. As ZOCs can overlap and are added this can lead to a loss of readiness up to 50%.
- **Shattered/scuttled:** If a unit is forced to retreat but there is no hex without enemy ZOC and the unit has a readiness below 50% or cannot retreat due to lack of fuel, it will be considered as "shattered" and will be removed from the game. Sea units that share that fate are labeled as "scuttled".
- **Destroyed:** If the strength of a unit falls to 0 due to a battle, it is destroyed and will be removed from the game.
- **Surrenders:** If one of the following cases occurs:
 - combat readiness of the defender is lower than 50% and the strength of the unit is three or lower
 - the unit is out of ammo, cannot retreat and its strength is five or lower
 - the unit is out of ammo and its strength is three or lower

Then an attack on the unit leads to its surrender and it is removed from the game.

10.5. Support fire

Fighter, artillery and anti-air units can fire support shots if certain circumstances are met. If there is a battle within the range of such a unit it will participate automatically.

This support fire only occurs once during a single battle, but may occur in a number of different battles as long as the unit is supplied with ammunition.

- **Artillery** intervenes if friendly ground troops are attacked by enemy ground troops.
- **Anti-air** intervenes if friendly ground troops are attacked by enemy aircraft.
- **Fighter** intervene if friendly aircraft are attacked by enemy aircraft.

In such a battle, the supporting unit cannot lose strength points. Units that delivered support fire are marked on the map. This will also happen if the unit originally is hidden in the fog of war. The marker will be removed at the end of the turn.

10.6. Combat results

The result of a battle is displayed in the combat window where all factors are listed if the player has chosen detailed combat info.

All combats and message of a scenario are stored and saved with a savegame. They can be viewed in the combat protocol where you can get by pressing "C" key or by clicking the "i" button on the panel and then click "combats" in the next menu.

11. Experience

Units gain experience by damaging enemies, capturing dense urban areas surviving a battle without losses and by destroying an enemy unit. The current experience points of a unit can be seen in the headquarter or in the panel if the unit is selected.

By collecting experience points the unit can reach new levels that improve their combat results.

Levels, experience and the impacts are listed here:

Level	Experience needed	Battle modifier
0	0	1
1	1000	1.3
2	2000	1.6
3	3500	1.9
4	5000	2.2
5	7000	2.5

11.1. Reinforcement and experience

If a unit receives reinforcements to compensate losses endured in combat, its experience will decrease by 5% for each new strength point due to the inexperienced new recruits.

Example: A unit with 100 experience points receives 5 new strength points. This results in a decrease to 75 experience points ($100 \cdot (100\% - 5 \cdot 5\%)$).

12. Support units (SUs)

Support units or short SUs are units that are subordinates of core or auxiliary troops. Which and how many SUs are available depends on the unit type and the formation of a unit. The costs vary as they include new strength that considers the equipment of the main unit.

This table lists all SUs and their attributes:

SU	Type	Formations	Strength	Changes
Additional battalion	All ground	Div.	+15	
Additional company	All ground	Reg.	+15	
Additional squadron	All air	All	+15	
Motorised antitank company	Tank	All	+5	Defense against tanks +3
Selfpropelled antitank company	Tank	All	+5	Attack on tanks +3
Rifle company	Tank	All	+5	Urban battles +5
Tank recovery platoon	Tank	Div.	-	Strength cost -10%
Flame tank platoon	Tank	Div.	+5	Ignores entrenchment up to one level as well as "Bunker bonus"
Antitank company	Infantry	All	+5	Defense against tanks +3
MG company	Infantry	All	+5	Defense against infantry +2
Mortar company	Infantry	All	+5	Against infantry: attack +2, defense +1
Recon platoon	Recon	All	+5	View +1
Heavy mechanized company	Recon	All	+5	Against armoured vehicles: Attack +1 Defense +3
Special supply platoon	Artillery	All	+5	Against armoured vehicles: Defense +4
Heavy antitank platoon	Antitank	All	+5	Against armoured vehicles: Attack +3 Defense +3
Heavy antiair platoon	Antiair	All	+5	Against aircraft: Attack +2 Defense +3
Fighter squadron	Bomber	All	+5	Against aircraft: Defense +4
Dive bomber squadron	Bomber	All	+5	Against armoured vehicles: Attack +4
Recon squadron	Fighter, Bomber	All	+5	View +1
Observation company	Artillery	Div., Reg.	+5	Against ground targets +1
Antiair company	Infantry,	Div., Reg.	+5	Against aircraft:

	Tank, Antitank, Artillery			Attack +2 Defense +3
Supply company	All ground	All	+5	Ammunition +1, Fuel+5

How to assign support units see 13.4.

13. Officers

Officers are an important element of a unit. They can have a decisive influence on defeat or victory and shape the career of a formation.

Each officer has three basic values, some have a fourth value. That special value increases the effect of an officer in certain situations.

All values have six levels that change by time. Each action and each battle influences the levels as the officer gains experience for each value.

13.1. Leadership

Units lose readiness points e.g. by fighting or movement. An officer helps regenerate readiness. How much he can boost it depends on his leadership level. Victories against enemy units increase leadership, defeats or even the loss of a unit lower the leadership value of an officer.

How the levels influence readiness:

Level	Readiness bonus/Turn
0	+3
1	+6
2	+9
3	+11
4	+13
5	+15

13.2. Attack

The current attack value of an officer is added to the attack value of his unit thus he enforces its attack. Each successful attack increases this value gradually.

Attack levels and their influence:

Level	Attack bonus
0	0
1	+2
2	+4
3	+5

4	+6
5	+7

13.3. Defence

The current defense value of an officer is added the defense value of a unit and thus he bolsters its defense. Each successful defense is increasing this value gradually.

Defense levels and their influence:

Level	Defence bonus
0	0
1	+2
2	+4
3	+5
4	+6
5	+7

13.4. Specialities

There are three kinds of specialities: specialized against a combat type, supply and reinforcement.

13.4.1. *specialized against a combat type*

Some officers have a notably talent to fight a certain type of enemy on the battlefield. The value will be activated and added each time the unit under the officers command is fighting against this special type of enemy. Such a combat type can e.g. be tanks. This means every time the unit fights against a tank unit its will have better combat results under this officer.

Each successful battle against the combat type the officer is specialized in will increase this value.

How the levels influence the battle:

Level	Special bonus
0	+0
1	+1
2	+2
3	+3
4	+4
5	+5

13.4.2. *Reinforcement*

Officers with that attribute make regaining strength for a unit cheaper. The higher this level is the lesser the costs of rebuilding its strength.

Every time the unit has to be reinforced, this value increases gradually.

How the levels influence the costs:

Level	Reduces costs by
0	0
1	-10%
2	-20%
3	-30%
4	-35%
5	-40%

14. ★ Headquarter



You can get to the headquarter by pressing the respective button on the panel or by using the key "h". In the HQ you can improve your units before the deployment or form new units.

14.1. Unit overview

At the beginning of a mission your core units will be displayed in the headquarter. They are marked with a golden badge. Which kind of troops will be shown in the list can be decided by choosing the filters on the right of the list (Core units, auxiliary units, reserve, all). The last selection is kept until the player changes it again. More information on units see chapter 6. Units.

14.2. Forming new units

Forming of new units takes place in the headquarter. The button "New" has to be active for this. If it is inactive, hovering over the button will show a tooltip with information about why it is not possible to form new units. This may happen if there are no deploy hexes or unit slots left. Are all necessary conditions met, you advance to the recruitment. Here you have to choose which unit type you wish to form and then select the predominant equipment of the unit to be. On the left you can choose an officer, if there is more than one in the pool. On the right you choose your formation either division or regiment. The costs of the selections are updated with each change

and displayed in the lower middle of the screen. If all is set as the player wishes, he can buy the unit by pressing "Recruit". The unit will now be transferred to the reserve. Now the unit can be placed on the map by clicking on "Deployment". This will only be possible if there are hexes to deploy the unit. Note that for deployment the same rules apply as for movement, which means units cannot be placed on hexes they normally could not enter. As soon as a unit is deployed it can move and fight.

14.3. Disband a unit

The currently selected unit in the headquarter can be disbanded, if it is not adjacent to an enemy. In that process you receive 50% of the current value in resources. If you hover over the "Disband" button you will get a tooltip with a preview of the amount. Auxiliary troops may be disbanded but the player receives no resources for them.

14.4. Improving units

As long as a unit is in the reserve it can be improved by exchanging or adding elements. For this you have to select the unit in the hq and push the "Reequip" button. In the follow up menu you can select the elements discussed in the next paragraphs:

14.4.1. Equipment

The main equipment of a unit provides the basic combat values of it. Exchanging it can increase them. You have to keep in mind that there are improvement trees for some marks. For example it is much cheaper to advance a Panzer III to the next generation of its kind than switching to the Panzer IV line. Experience is not touched by this action.

14.4.2. Support units (SUs)

You can add smaller sub-unit to a unit that are called support units or short SUs. A regiment can controll one such unit, but a division two.

Adding an SU costs resources that are different for each of them. SUs change the basic values of a unit according to its nature. Costs and change in the values are shown before the purchase. All details on the different SU can be studied in chapter 12. Support units.

14.4.3. Officers

Each unit has an officer that can be exchanged at the costs of resources. The amount depends on the abilities of the officers which means the better the officer the higher the costs of assigning him to a unit.

Officers can only be assigned if they are in the officers corps allowed in scenario. This may differ from map to map and each officer as it is related to the design of the scenarios. You cannot form new units without officers.

14.4.4. Transports

Infantry, artillery, antitank and antiair can be equipped with transports. This has a huge impact on the range of such units.

Transport can also be removed or exchanged. Details on the various transports can be seen in 8.2. Transports

14.4.5. Formations

There are two kinds of formations in Army General, divisions and regiments. This are their differences:

	Regiment	Division
Strength points	50	100
SU slots	1	2
Additional costs for reinforcements	0%	25%
Combat factor	50%	100%

15. Weather

Weather has strong influence on actions of the units. All units of a certain type are effected likewise (e.g. all air units). At the beginning of a turn, the weather is set for the whole map. Here os listed which kind of weather influenced what kind of units and how:

Weather Influence

Clear No restrictions

Cloudy View air units -25%

Air combat -20%

Air-ground or ground-air attacks -50%

Nebel Air-ground or ground-air attacks -50%

View ground units -50%.

Rain Range air units -30%

View air units -50%

Air combat -70%

Air-ground or ground-air attacks -70%

Range ground units -30%

Snow Range of air units -50%

View air units -50%

Air combat -70%

Air-ground or ground-air attacks -70%

Range ground units -20%

View ground units -50%

16. Victory and defeat

Victory and defeat are decided by the mission objectives. They are documented for both sides in each scenario and can be viewed via the objective screen anytime. If a player reaches his mission goals sooner than his enemy he wins the battle. He loses the scenario in the opposite case. During a campaign this might lead to a different branch or outcome depending on the design.

17. End of a turn

At the end of these things happen:

- the entrenchment level of all units that did neither fight nor have been moved is increased by one unless it has already reached the maximum level possible on the hex.
- Used action and movement points are restored again
- Determined resources are given to the players
- Symbols of support fire are removed from the map
- Combat readiness of all units increases according to the map settings and their officers values.

18. The campaigns

Campaigns are a chain of scenarios. Those again depict single battles. Even though most battles have a historic background, there are fictional maps as well.

During a campaign the player usually has core units. Between scenarios they can be refitted and improved. If there is no limit caused by unit slots the player can form new units if enough resources are available.

18.1. Deployment

At the beginning of a new map you often can place parts or even all your core units on the map. This is only allowed on special hexes that are marked. There is also a distinction between ground, sea and air units. If there are markers at sea, you can place ships or ground troops which will be turned into sea transports there. This may cost resources depending on the scenario. The same way ground troops can be placed on air deploy hexes as an air transport if the unit is capable of being transported this way. If there are two options you have to click the upper half of the hex for the air option, the lower half if you wish to deploy on the ground or sea.

18.2. Operation Compass

After the failed Italian attempt to conquer Egypt, Britain is gathering troops commanded by A. Wavell for a counter strike. The operation, originally planned as a five day raid, is quickly expanded due to the huge successes in its early stages. Drive the Italian forces back to El Agheila in Libya and destroy the 10. Army of the enemy.

18.3. Mare Nostrum

Great Britain's success during Operation Compass leads, after initial amusement on the German side, to the sobering insight that Africa might soon be completely in Allied hands if Germany stays on the sideline. Therefore the Afrikakorps is formed and sent away to stop the British. The events in Europe also changed the situation in Libya when the Germans arrive. Exploit the current weakness of the Allies to regain lost territory. Maybe you can even achieve a breakthrough into Egypt and the Suez channel ?

19. Map editor

At the start you only see the menu bar of the editor and a random map. Each menu entry shows a window providing more options that are discussed in the next paragraphs.

Right of the bar there are two buttons, first the switch to toggle air and ground view and next to it the button you can change the nation you edit.

Which nations can be chosen on that button can be set in the scenario settings.

19.1. Options

19.1.1. *Main menu*

Return to the main menu. Maps that were not saved will be lost.

19.1.2. *Quit game*

Return to the desktop. Maps that were not saved will be lost.

19.1.3. *Save*

Saves the current state of the map by the name seen in the input field below the button. An older state of the map bearing the same name will be overwritten without warning.

19.1.4. *Load*

A map has to be located in the folder of the editor and be selected from the list below the button. The filename will show up in the input field and the map can be loaded. The folder is located at a directory that is something like this:

C:\Users\Name\Documents\Army General\Scenarios

19.2. Landscape

The landscape of a hex can be designed in two ways: You can set all elements and then imprint them on hexes with "field" as a whole. Or you select each component for each hex individually.

19.2.1. *Hexes*

Field: This selection imprints all settings listed below on a hex at once.

Terrain: Here you can select if the hex forms a hill, a river and so forth. Each type influences movement, entrenchment and combat. More about the values see 5.1.

Ornaments: Important elements that can influence the game can be set here. Railroads, streets and swamps are only a few of them. They also can influence the terrain type (e.g. forest)

Objects: Urban areas, arifields and other facilities and be put on the map with this function. It will also open the attributes menus for the objects. see next paragraph and chapter 5.2.

19.2.2. *Attributes of objects*

Name: Any name can be chosen for an object here.

Owner: The drop down menu contains the both nations that were selected for the map by the designer.

Supply field: If this option is active the owner of the object can place new units on the hex and its adjacent fields.

19.2.3. *Special*

Deploy hex: During a campaign the player can place core units at the beginning of a new mission. Deploy hexes can only be used once for this purpose. With this function you can place them on the map.

Supply hex: These hexes act like deploy hexes but they can be used infinitely.

19.3. Units

In the unit menu there is a column of buttons each of which represents a weaponry branch in the game. If you click on one of those the list left of the buttons will change and show all possible equipments of that branch from all nations. The content of the list depends on the starting date of the scenario as only weapons will be displayed that were available up to that date. As the weapons of all nations are shown regardless of the set nation you can assign equipment to a nation that originally did not produce it e.g. a Soviet T-34 can be assigned to German unit.

The values of the equipment is displayed below the list. You can also assign transports to the unit there.

If you select an equipment in the list you can then place it on the map with left clicks. This also opens a window with attributes of the unit, the designer can change. If you enter values that exceed the maximum possible, they will be capped by the editor:

Name: By default the name of the equipment is written here. The designer may choose better once like historical unit names. This is only for atmosphere and does not influence anything.

Strength: By default the maximum strength is set here. You can depict weaker units or ones that suffered losses before by entering lower values. One strength point is the minimum amount that needs to be entered.

Ammunition: By default the maximum is set. You can set any value from 0 to the maximum amount. Units without ammo suffer additional losses in a battle.

Experience: Values up to 7000 are sensible in this field. Note the levels of experience and their impact discussed in chapter 11.

Fuel: Contains the maximum amount. Values ranging from 0 to the start value can be entered.

Entrenchment: The level of entrenchment depends on the unit and the terrain. It can only be as high as the lower of these two values.

Core unit: This is important for campaigns as this kind of unit can be transferred to new missions. In single scenarios this option can be used in combination with limited unit slots to control the number of units the player can use.

19.4. Scenario settings

Here rough settings of a scenario can be adjusted.

19.4.1. *Map*

Map size: The size in number of hexes is set here. Changes will only be applied if you press "new map". Note that this will reset the map and all elements are wiped from the map. Old maps will not be expanded but just converted to an empty map with a different size. If you save your map you might want to change the filename for that reason.

Date: The date influences the equipment that is available for creating the map and for the player to buy. Weapon systems are only given if their historical production start is reached. Weapons that already have reached the end of their production cannot be bought in the game, but put on the map in the editor. Such units can still be used and refitted by the player.

19.4.2. *Player*

The settings for the player contain:

Nation: Which country is on one of the sides. Also changes the set weapons and officers the player gets to play with.

AI: You can choose aggressive AI, defensive AI or human player.

Team: Which block is your nation part of ? You can choose Western Allies, Axis or Sowjet Union.

Resources at start: If the scenario is started, this amount of resources will be added to the players resources pool.

Resources per turn: The set number will be added to the players resources pool at the beginning of each turn.

19.4.3. *Weather*

Weather has a huge impact on an military operation. In this menu you can set the chance for each kind of weather on a map. The values range from 100% (each turn this kind of weather will take place) to 0% (it never occurs). The sum of all percentages distributed should equal 100 in the end.

20. ★ Credits

Army General was created by:

PHOBETOR

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Ronald Wendt

LEAD PROGRAMMER

Thomas Gereke

GRAPHICS

Riko Jensen

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BETA TESTERS

We used the following libraries to create Army General:

Parts based on code of the game

'Operation Barbarossa - The Stuggle for Russia'

coded by Ralf Zenker

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Ronald wants to thank some people without whom this game would not have been possible :

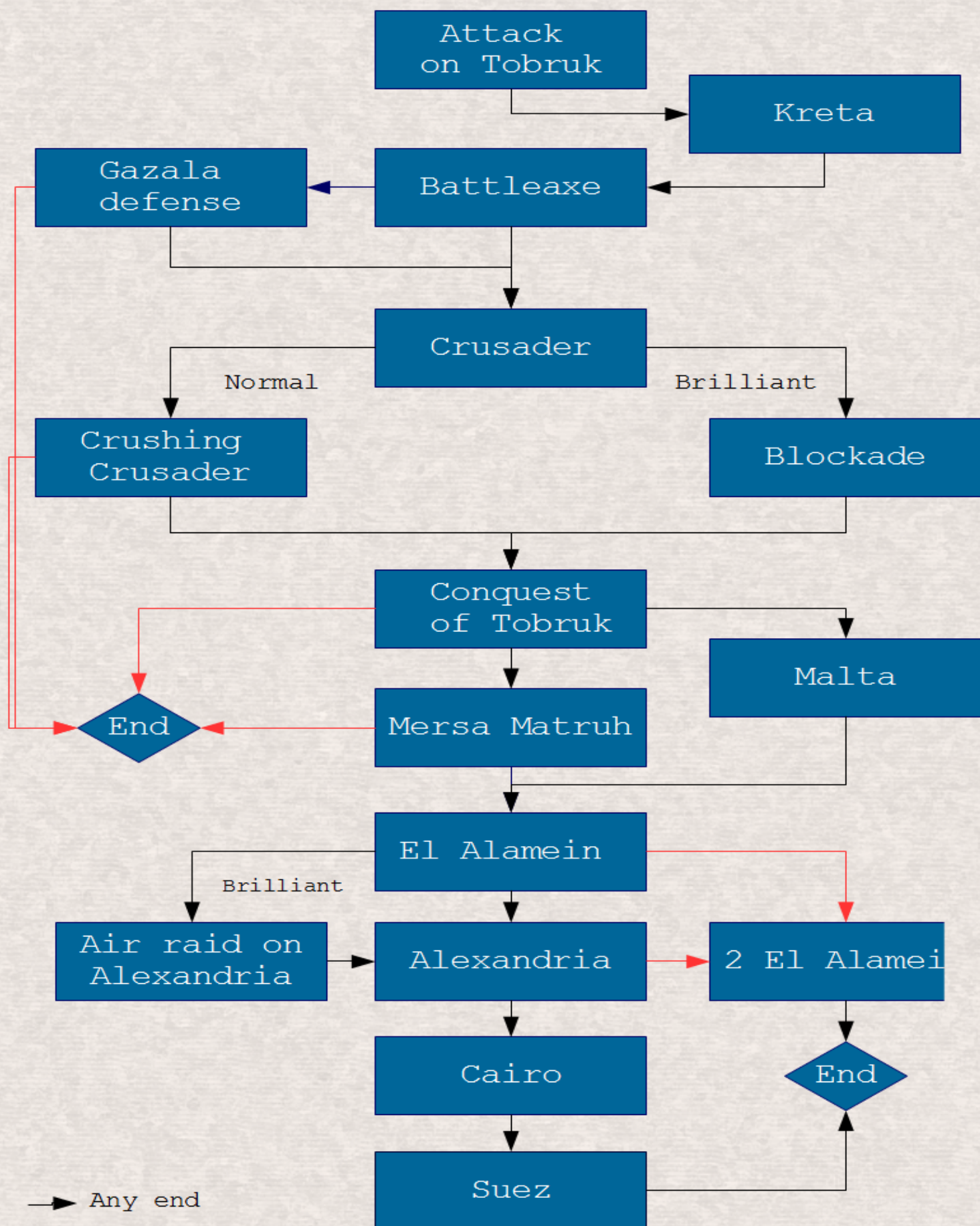
Klaus and Luise Wendt for your everlasting support

Keiko Wendt for being patient with a busy father

Christiane Wendt for being a good friend when its really needed

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→ Any end



Operation Compass

